FX Carry Trade: Portfolio Construction, Crash Risk and Factor Analysis

Eight G10 currencies against the US dollar, August 2016 – August 2026

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Abstract

We test a monthly long-short carry strategy on eight major currencies against the US dollar (EUR, GBP, JPY, CHF, NOK, AUD, NZD, CAD) using Bloomberg spot and one-month forward data, September 2016 to July 2026 (119 months). Buying the three highest-rate and selling the three lowest-rate currencies made 2.92% a year after costs, with a Sharpe ratio of 0.44. The profit came almost entirely from the interest-rate gap; exchange-rate moves added nothing, against uncovered interest parity (UIP). Returns are skewed to the downside, and the five worst months all match global shocks (COVID-19, the Fed hikes of 2018 and 2022, Omicron and the 2024 yen-carry unwind). Carry does well in calm markets, loses in crises and moves with equities; after controlling for equity and volatility risk its extra return is close to zero. A three-month momentum filter made results worse. Carry is a reward for crash risk, not a free lunch.

1 Introduction

A carry trade borrows in a low-rate currency and invests in a high-rate one. Under uncovered interest parity (UIP) the high-rate currency should fall by the rate gap, leaving no profit. In practice it usually does not fall enough, so carry makes money over long periods (Fama, 1984; Engel, 1996; Lustig et al., 2011). The risk is sudden crashes, such as the yen-carry unwind of 2008: carry goes "up the stairs and down the elevator" (Brunnermeier et al., 2009).

The period 2016–2026 is a good test. It covers the Fed hikes of 2017–18, the COVID-19 crash, the 2022–23 global rate rises, the end of negative rates in Japan and the August 2024 yen unwind, the April 2025 tariff shock and the 2026 oil shock. We ask: (1) Does carry earn a positive Sharpe ratio, and how does it vary between calm and stressed markets? (2) Does UIP hold, and where does the return come from? (3) How large is crash risk, and which events caused it? (4) Is carry a risk factor linked to equities and the VIX? (5) Does a momentum filter help?

2 Literature Review

UIP and its failure. Under UIP, the forward rate should predict the future spot rate. Fama (1984) regressed spot changes on the forward premium and found slopes below one and often negative (the *forward premium puzzle*); Hansen and Hodrick (1980) and Engel (1996) confirm this across currencies. Since 2008, covered-parity gaps also affect JPY and CHF forwards (Du et al., 2018).

Carry as a risk premium. Lustig et al. (2011) show that a dollar factor (DOL) and a carry factor (HML_{FX}) explain most currency returns, with high-rate currencies loading positively on

carry. Menkhoff et al. (2012a) find carry loses when global FX volatility rises, while JPY and CHF hedge. Koijen et al. (2018) find carry across asset classes, and Burnside (2012) reviews the risk-based explanations.

Crash risk. Brunnermeier et al. (2009) show carry is negatively skewed and crashes when the VIX rises and traders unwind. Jurek (2014) finds option hedges remove part of the return; Burnside et al. (2011) point to rare unseen disasters; Daniel et al. (2017) document large drawdowns. Menkhoff et al. (2012b) find momentum weak among major currencies. The BIS (Bank for International Settlements, 2025, 2024) documents the size of FX funding markets and links August 2024 to the unwinding of yen carry.

3 Data and Methodology

3.1 Bloomberg data

GP: daily spot rates for EURUSD, GBPUSD, AUDUSD, NZDUSD, USDJPY, USDCHF, USDCAD, USDNOK, plus the S&P 500 and VIX (18 Aug 2016 – 17 Aug 2026, 2,608 days). **FRD**: one-month forward points (forward = spot + points/10,000; /100 for JPY). **WIRP**: implied central-bank paths on 17 Aug 2026, used as a forward-looking check only. **FXIP**: forward points grid; positioning data were not available. All rates are quoted as USD per foreign currency, so a positive return means the currency rose against USD. We use month-end data and drop the partial August 2026, leaving 119 holding months.

Table 1: The eight currencies against the US dollar, September 2016 – July 2026

Currency	Rate gap vs USD (% p.a.)	Spot change (% p.a.)	Carry return (% p.a.)	Volatility (% p.a.)	Months long (%)	Months short (%)
EUR	−1.92	+0.57	−1.36	6.92	0	98
GBP	−0.68	+0.57	−0.11	7.90	44	1
JPY	−2.86	−3.82	−6.67	9.03	0	99
CHF	−3.02	+2.27	−0.76	7.47	0	100
NOK	−0.59	−0.61	−1.20	11.78	53	0
AUD	−0.40	−0.23	−0.62	9.68	53	0
NZD	−0.13	−1.54	−1.67	10.72	82	0
CAD	−0.64	−0.47	−1.11	6.47	68	2

Rate gap = one-month forward discount, annualised (roughly foreign rate minus US rate). Carry return = return from holding a one-month forward and rolling it each month.

Over this period the US dollar paid more than every other currency (Table 1). CHF (−3.0 points), JPY (−2.9) and EUR (−1.9) were the cheapest to borrow. Because the strategy ranks currencies against each other, the bought basket still paid 2.48 points more than the sold basket on average.

3.2 Portfolio construction

At each month-end the carry signal is the one-month forward discount:

$$c_{i,t} = \frac{S_{i,t}}{F_{i,t}} - 1 \;\approx\; \frac{i_{\text{foreign},t} - i_{\text{USD},t}}{12} \tag{1}$$

We buy the top three and sell the bottom three (one third each) and hold to the next month-end. The return of a forward position splits into a spot part and a carry part:

$$x_{i,t+1} = \frac{S_{i,t+1}}{F_{i,t}} - 1 \;=\; \underbrace{\left(\frac{S_{i,t+1}}{S_{i,t}} - 1\right)}_{\text{spot}} + \underbrace{c_{i,t}}_{\text{carry}} + \text{cross term} \tag{2}$$

Costs are 5 bp per unit of turnover. Benchmarks: a **dollar basket (DOL)** holding all eight currencies, and **static carry**, which keeps the first month's positions throughout.

3.3 Measures

Sharpe = mean $\times 12$ / volatility $\times\sqrt{12}$; Sortino uses downside volatility; Calmar = growth rate / maximum drawdown; tail ratio = average of the worst 5% of months / average of the best 5%. UIP is tested with the Fama (1984) regression $\Delta s_{i,t+1} = \alpha_i + \beta_i(f_{i,t} - s_{i,t}) + \varepsilon$, where UIP means $\beta = 1$. Regimes split months by the VIX median (17.5) and flag crisis months (VIX ≥ 30). The momentum filter keeps a long only if the currency rose over three months and a short only if it fell. Standard errors are Newey–West (3 lags).

4 The Long and Short Positions

Short side. The funding basket barely changed: CHF was sold in 100% of months, JPY in 99% and EUR in 98%. All three central banks kept rates at or below zero for most of the period. The only swaps were GBP for JPY in December 2021, and CAD for EUR from June 2026 after the ECB hiked.

Long side. The long basket rotated as central banks diverged:

- **2016–2019: AUD, NZD, CAD** — highest rates at the start; held two years (July 2017 – June 2019).
- **2019–2022: NOK, NZD, CAD** — frequent changes as COVID cuts squeezed all rate gaps.
- **2023 – mid 2024: GBP, NZD, CAD** — the Bank of England hiked to 5.25%; held 19 months.
- **Dec 2024 – 2026: AUD, GBP, NOK** — the RBA and Norges Bank kept rates high and hiked again in 2026.

Every position is against USD, so the dollar legs largely cancel and the portfolio is in effect long crosses such as NZD/JPY, CAD/CHF and GBP/EUR (Tables 2 and 3).

Table 3: Position history: every period in which the long and short baskets stayed the same

From	To	Months	Long (high-rate)	Short (funding)	Return (%)
2016-09	2017-02	6	AUD, NOK, NZD	CHF, EUR, JPY	+6.85
2017-03	2017-03	1	AUD, CAD, NZD	CHF, EUR, JPY	−1.59
2017-04	2017-06	3	AUD, NOK, NZD	CHF, EUR, JPY	−0.26
2017-07	2019-06	24	AUD, CAD, NZD	CHF, EUR, JPY	−2.14
2019-07	2020-03	9	CAD, NOK, NZD	CHF, EUR, JPY	−10.33
2020-04	2020-04	1	AUD, NOK, NZD	CHF, EUR, JPY	+3.94
2020-05	2020-06	2	AUD, CAD, NZD	CHF, EUR, JPY	+3.53

From	To	Months	Long (high-rate)	Short (funding)	Return (%)
2020-07	2020-07	1	CAD, NOK, NZD	CHF, EUR, JPY	−0.26
2020-08	2020-09	2	AUD, CAD, NZD	CHF, EUR, JPY	+0.73
2020-10	2020-10	1	AUD, NOK, NZD	CHF, EUR, JPY	−1.50
2020-11	2021-05	7	CAD, NOK, NZD	CHF, EUR, JPY	+11.20
2021-06	2021-08	3	CAD, GBP, NZD	CHF, EUR, JPY	−1.58
2021-09	2021-11	3	CAD, NOK, NZD	CHF, EUR, JPY	−0.16
2021-12	2021-12	1	CAD, NOK, NZD	CHF, EUR, GBP	+0.43
2022-01	2022-02	2	GBP, NOK, NZD	CHF, EUR, JPY	+0.22
2022-03	2022-03	1	CAD, NOK, NZD	CHF, EUR, JPY	+3.99
2022-04	2022-04	1	GBP, NOK, NZD	CHF, EUR, JPY	−0.46
2022-05	2022-05	1	CAD, GBP, NOK	CHF, EUR, JPY	−0.56
2022-06	2022-06	1	CAD, GBP, NZD	CHF, EUR, JPY	−0.58
2022-07	2022-09	3	AUD, CAD, NZD	CHF, EUR, JPY	−2.33
2022-10	2022-10	1	CAD, NOK, NZD	CHF, EUR, JPY	+4.65
2022-11	2022-11	1	CAD, GBP, NZD	CHF, EUR, JPY	−1.05
2022-12	2022-12	1	CAD, NOK, NZD	CHF, EUR, JPY	−3.11
2023-01	2024-07	19	CAD, GBP, NZD	CHF, EUR, JPY	+7.10
2024-08	2024-11	4	GBP, NOK, NZD	CHF, EUR, JPY	+1.16
2024-12	2026-05	18	AUD, GBP, NOK	CHF, EUR, JPY	+11.16
2026-06	2026-07	2	AUD, GBP, NOK	CAD, CHF, JPY	+0.35

Return = compounded net return of the portfolio over the period.

The two longest holding periods were the most profitable (+7.1% and +11.2%); the worst was July 2019 – March 2020 (−10.3%), long NOK, NZD and CAD into COVID and the oil collapse.

5 Results: Portfolio Performance

Main result. Carry made 2.92% a year after costs, with 6.61% volatility, Sharpe 0.44 and Sortino 0.64 (Table 4). It was positive in 59% of months, but the t-statistic of 1.79 is significant only at 10%. The maximum drawdown was 14.0% (August 2017 – March 2020, recovered March 2021). Costs are negligible: only 8% of the portfolio is traded per month.

Against buy-and-hold. Rebalancing added value. Static carry made 1.76% (Sharpe 0.23) because it held NZD and AUD after their rates fell. The dollar basket lost 1.69% (Sharpe −0.23) against a strong, high-yielding dollar. The S&P 500 made 13.7%, but carry rises and falls with it (Section 8), so it does not diversify equities.

Source of return. The carry part gave 2.48% a year (t = 14.2); the spot part gave 0.49% (t = 0.30), i.e. nothing (Table 5, Figure 2). The strategy earned the rate gap and exchange rates did not take it away.

Over time. Before 2023 the Sharpe ratio was about 0.2 (Table 6, Figures 3 and 4): 2018 lost 4.6% on Fed hikes and the US–China trade war, and 2020 was slightly negative. From 2023 carry made 5.39% a year (Sharpe 1.24) as Japan and Switzerland stayed near zero while the UK, Norway and Australia kept rates high. January–July 2026 made 7.6%.

Calm versus stressed markets. Carry is a good-weather strategy (Table 7, Figure 5): Sharpe 0.64 when the VIX is low, 0.32 when it is high, and −0.13 in crisis months. But months that *start* with a high VIX earned 5.94% a year (Sharpe 0.82), because they include the rebounds after sell-offs. Holding carry through fear was rewarded, as a risk premium should be.

Table 2: How each currency position is traded, and how often it was used

Currency	Bloomberg pair	Months long (%)	Trade when long	Months short (%)	Trade when short
NZD	NZDUSD	82.4	Buy NZDUSD	0.0	Sell NZDUSD
CAD	USDCAD	68.1	Sell USDCAD	1.7	Buy USDCAD
AUD	AUDUSD	52.9	Buy AUDUSD	0.0	Sell AUDUSD
NOK	USDNOK	52.9	Sell USDNOK	0.0	Buy USDNOK
GBP	GBPUSD	43.7	Buy GBPUSD	0.8	Sell GBPUSD
EUR	EURUSD	0.0	Buy EURUSD	98.3	Sell EURUSD
JPY	USDJPY	0.0	Sell USDJPY	99.2	Buy USDJPY
CHF	USDCHF	0.0	Sell USDCHF	100.0	Buy USDCHF

Long a currency = buy it against USD; short = sell it against USD. For pairs quoted as USD per currency (USDJPY, USDCHF, USDCAD, USDNOK) the trade direction is reversed.

Table 4: Performance summary, monthly returns, September 2016 – July 2026 (119 months)

	Carry (net)	Carry (gross)	Static carry	Dollar basket	S&P 500
Average return (% p.a.)	2.92	2.97	1.76	−1.69	13.73
Growth rate, CAGR (%)	2.73	2.78	1.49	−1.93	13.30
Volatility (% p.a.)	6.61	6.61	7.53	7.26	15.40
Sharpe ratio	0.44	0.45	0.23	−0.23	0.89
Sortino ratio	0.64	0.65	0.33	−0.31	1.42
Maximum drawdown (%)	−14.04	−14.03	−17.13	−25.67	−24.77
Calmar ratio	0.20	0.20	0.09	−0.07	0.54
Skewness	−0.54	−0.54	−0.51	0.03	−0.43
Excess kurtosis	0.99	1.00	0.47	−0.34	0.62
Worst month (%)	−6.41	−6.41	−6.84	−5.25	−12.51
Best month (%)	4.65	4.67	4.85	5.55	12.68
Positive months (%)	58.8	58.8	59.7	43.7	67.2
Tail ratio	1.08	1.08	1.22	1.04	0.95
t-statistic (mean = 0)	1.79	1.83	0.92	−0.82	3.45

Net = after 5 bp per unit of turnover. Static carry keeps the September 2016 positions. Dollar basket = equal weights in all eight currencies against USD. S&P 500 = price return.

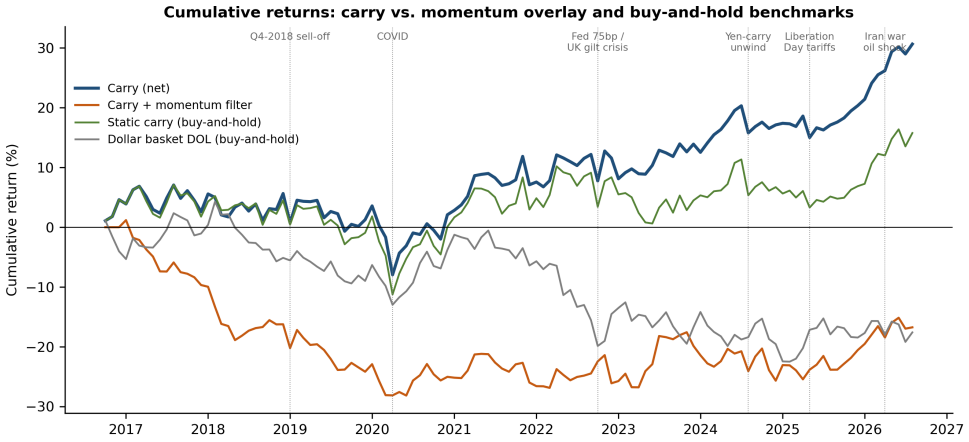


Figure 1: Cumulative returns of carry, carry with the momentum filter, and the two buy-and-hold portfolios. Dotted lines mark the main market shocks.

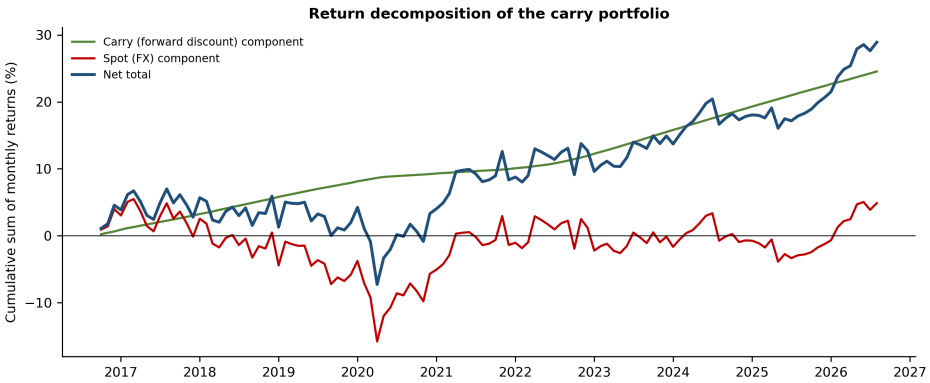


Figure 2: Cumulative carry part and spot part of the portfolio return.

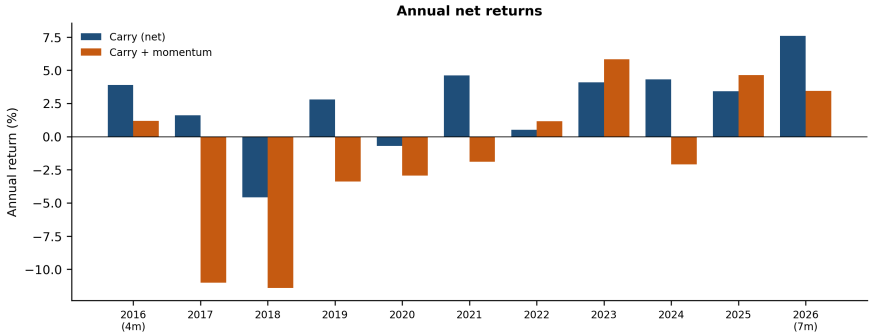


Figure 3: Yearly net returns of carry and carry with the momentum filter (2016: Sep–Dec; 2026: Jan–Jul).

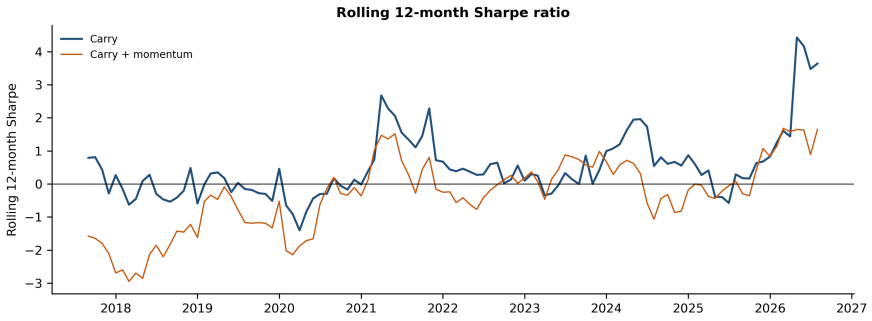


Figure 4: Rolling 12-month Sharpe ratio.

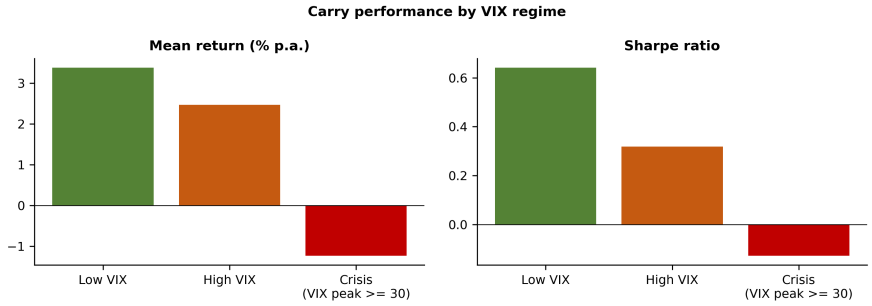


Figure 5: Average return and Sharpe ratio of carry in low-VIX, high-VIX and crisis months.

Table 5: Where the return comes from

Part of return	Average per month (%)	Per year (%)	t-statistic
Spot (FX) component	0.041	0.49	0.30
Carry (forward discount) component	0.206	2.48	14.25
Cross term	0.000	0.00	0.18
Gross excess return	0.247	2.97	1.83
Transaction costs	0.004	0.05	4.44
Net return	0.243	2.92	1.79

Table 6: Performance by sub-period (net)

Period	Months	Return (% p.a.)	Volatility (% p.a.)	Sharpe	Max drawdown (%)	t-stat
2016-09 to 2019-12	40	1.27	6.68	0.19	−7.23	0.51
2020-01 to 2022-12	36	1.79	8.61	0.21	−8.24	0.43
2023-01 to end	43	5.39	4.34	1.24	−4.42	2.76

Table 7: Carry performance in calm and stressed markets

Market regime	Months	Return (% p.a.)	Volatility (% p.a.)	Sharpe	Worst month (%)	Positive months (%)
Low VIX (below median 17.5)	59	3.37	5.26	0.64	−3.78	61.0
High VIX (above median)	60	2.47	7.75	0.32	−6.41	56.7
Crisis (VIX $\geq$ 30 in month)	25	−1.23	9.63	−0.13	−6.41	48.0
VIX low at start of month	59	−0.16	5.77	−0.03	−4.25	54.2
VIX high at start of month	60	5.94	7.29	0.81	−6.41	63.3

Regimes use the average VIX in the month, or the VIX at the start of the month. Crisis months: 25, mostly in 2020 and 2022.

6 Results: UIP and the Forward Premium Puzzle

Table 8: UIP test (Fama regression), monthly data, annualised

Currency	β	Std. error	t ($\beta = 0$)	t ($\beta = 1$)	p-value ($\beta = 1$)	R^2	Months
EUR	1.71	2.84	0.60	0.25	0.801	0.003	119
GBP	0.65	3.67	0.18	−0.09	0.925	0.000	119
JPY	1.01	1.66	0.61	0.01	0.993	0.003	119
CHF	2.45	1.76	1.39	0.82	0.412	0.014	119
NOK	−3.01	4.17	−0.72	−0.96	0.337	0.002	119
AUD	−2.11	3.31	−0.64	−0.94	0.347	0.002	119
NZD	2.55	3.40	0.75	0.46	0.648	0.002	119
CAD	3.95	3.28	1.20	0.90	0.369	0.011	119
Pooled	1.15	1.61	0.71	0.09	0.926	0.004	952

UIP holds if $\beta = 1$; a random walk gives $\beta = 0$. Pooled = all currencies together with currency fixed effects and standard errors clustered by month.

Currency by currency. Slopes range from −3.0 (NOK) to 3.9 (CAD); pooled 1.15 (s.e. 1.61) (Table 8, Figure 6). No currency rejects UIP, but none rejects a random walk either, and R^2 is below 1.4%. Ten years of monthly data cannot settle the question: rate gaps moved with the US cycle, while exchange rates were driven by much larger shocks.

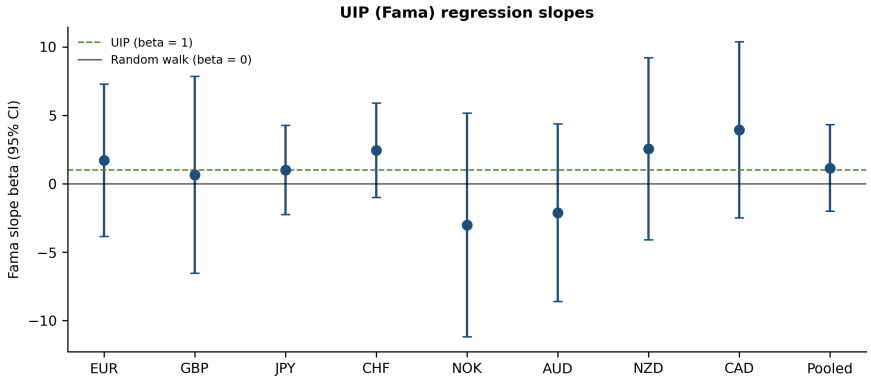


Figure 6: UIP slopes with 95% confidence intervals. UIP predicts $\beta = 1$ (dashed line).

Table 9: UIP at the portfolio level (% per year)

	Long basket	Short basket	Long minus short
Average rate gap vs USD (carry)	−0.13	−2.60	+2.48
Actual spot change vs USD	+0.26	−0.23	+0.49
Spot change UIP predicts	+0.13	+2.60	−2.48

Portfolio level. The portfolio test is clearer (Table 9). The sold currencies paid 2.6 points less than USD, so UIP says they should have risen 2.6% a year; they fell 0.23%. Long minus short, UIP predicts a spot loss of 2.48%; the actual result was a gain of 0.49%. The yen paid 2.9 points less than the dollar and still fell 3.8% a year as the Bank of Japan stayed near zero while the Fed went to 5.25–5.50%. This is the forward premium puzzle.

Implication. The carry return is the rate gap; the risk is in the spot part, which Sections 7 and 8 examine. Ranking on policy rates instead of forwards gives a similar result (Sharpe 0.35).

7 Results: Crash Risk

Shape of returns. Monthly skewness is −0.54 and excess kurtosis 0.99; daily figures are −0.74 and 8.2. The tail ratio is 1.08: the six worst months averaged −4.36% against +4.02% for the six best. The dollar basket, by contrast, is symmetric (skewness 0.03). The pattern matches Brunnermeier et al. (2009), although milder because G10 rate gaps were small.

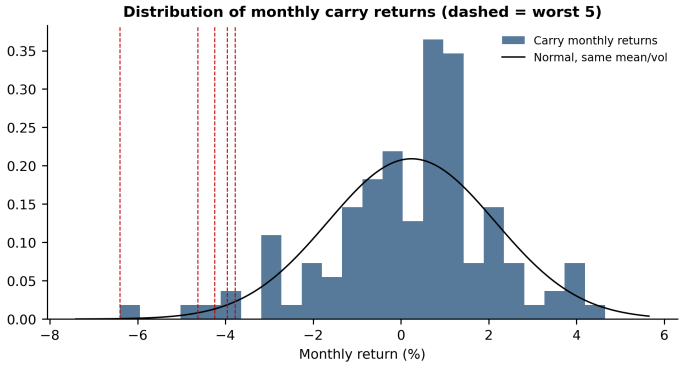


Figure 7: Monthly carry returns against a normal distribution with the same mean and volatility. Dashed lines mark the five worst months.

Table 10: The five worst months and what caused them

Month	Return (%)	Long side (%)	Short side (%)	Long / short	VIX	What happened
2020-03	−6.41	−6.35	−0.06	NOK, NZD, CAD / EUR, JPY, CHF	57.7	COVID-19 declared a pandemic; Saudi–Russia oil price war (9 Mar) hit NOK and CAD; rush into dollars; Fed cut rates to zero (15 Mar).
2018-12	−4.63	−2.82	−1.81	AUD, NZD, CAD / EUR, JPY, CHF	24.9	Fed hiked to 2.25–2.50% and kept shrinking its balance sheet (19 Dec); oil and stocks fell; US government shutdown (22 Dec).
2021-11	−4.25	−4.81	0.56	NOK, NZD, CAD / EUR, JPY, CHF	18.5	Omicron variant (26 Nov) caused an oil sell-off that hit NOK and CAD; Fed dropped the word “transitory” (30 Nov).
2022-09	−3.96	−6.67	2.71	AUD, NZD, CAD / EUR, JPY, CHF	27.3	Fed hiked 75 bp (21 Sep) and the dollar peaked; UK mini-budget crisis (23 Sep); Japan bought yen for the first time since 1998 (22 Sep).
2024-07	−3.78	−0.56	−3.22	GBP, NZD, CAD / EUR, JPY, CHF	14.4	Japan intervened to support the yen (11–12 Jul) and the BoJ hiked (31 Jul); the yen jumped and the short-yen leg lost; the unwind peaked on 5 Aug.

Long/short basket = contribution of each side to the monthly return.

Linking crashes to events. All five worst months were global sell-offs (Table 10, Figure 8), and all losses came from exchange rates, not carry. Two patterns:

- **Long basket fell** (Mar 2020, Dec 2018, Nov 2021, Sep 2022): NOK, NZD, AUD and CAD fell with oil and equities as the dollar rose.
- **Short basket rose** (Jul 2024): the yen jumped on the BoJ hike and Japanese intervention, forcing yen-funded traders to unwind at once — the 2008 pattern again.

Seven of the ten worst days were in March 2020 or April 2025 (Annex Table A2). In April 2025 the dollar fell, yet carry lost 3.0% because CHF, JPY and EUR rose even faster than the high-rate

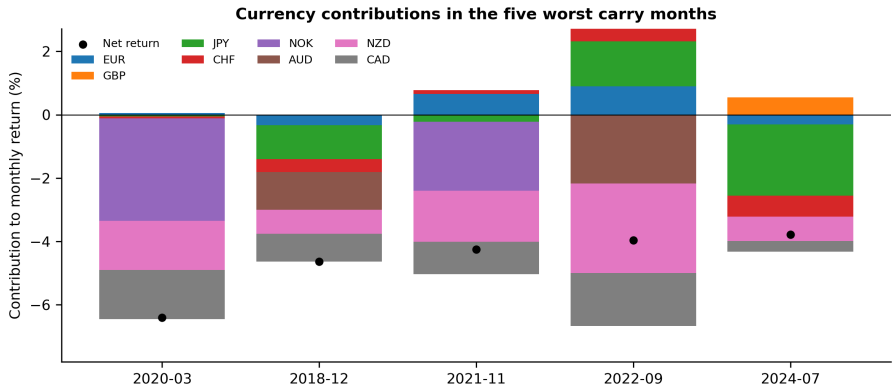


Figure 8: Contribution of each currency to the return in the five worst months (dots = total).

currencies: carry crash risk is about safe havens versus risky currencies, not the dollar. The 2026 Iran-war oil shock did not hurt (March 2026: +0.6%), since the strategy was long NOK and AUD and short oil importers JPY and EUR.

8 Results: Is Carry a Risk Factor?

Table 11: Correlation of carry with the S&P 500 and the VIX

Period	Monthly: S&P 500	Monthly: VIX change	Daily: S&P 500	Daily: VIX change
Full sample	0.56	−0.57	0.58	−0.52
2016-09 to 2019-12	0.53	−0.61	0.53	−0.43
2020-01 to 2022-12	0.72	−0.66	0.64	−0.56
2023-01 to end	0.20	−0.31	0.52	−0.53

Correlation over time. Carry moves with equities: correlation 0.56 with the S&P 500 and −0.57 with VIX changes (monthly; Table 11). The rolling one-year correlation (Figure 9) stayed between 0.30 and 0.79: about 0.75 after March 2020, 0.30–0.40 in calm 2023–24, and back near 0.7 after April 2025. It rises exactly when diversification is needed — the mark of a risk factor.

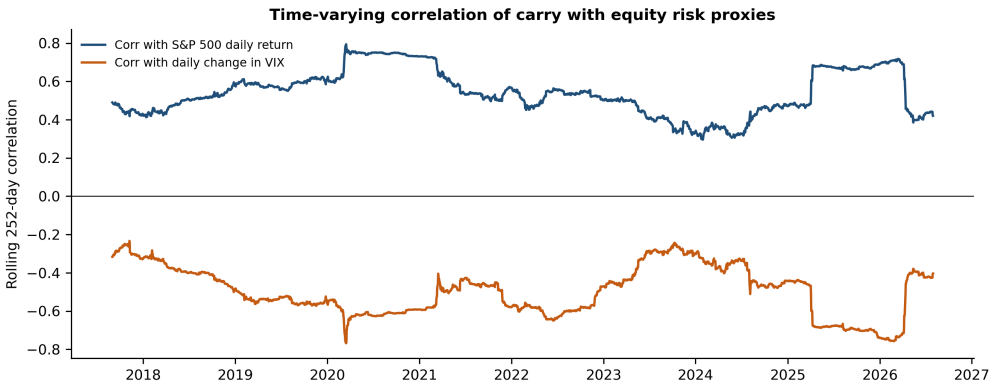


Figure 9: Rolling one-year (252-day) correlation of daily carry returns with the S&P 500 and with daily VIX changes.

Risk-adjusted return. The S&P 500 alone explains 31% of carry's variation ($\beta = 0.24$) and VIX changes 33% (−0.20% per VIX point); FX volatility shocks also hurt (Table 12). With both controls, the alpha falls from 0.24% to 0.10% a month and is insignificant (t = 0.73).

Table 12: Carry regressed on market risk (dependent variable: monthly carry return, %)

Risk measure	α (% per month)	t(α)	β	t(β)	R^2
S&P 500 return	−0.032	−0.22	0.240	6.24	0.31
Change in VIX	0.247	2.13	−0.199	−8.76	0.33
Global FX volatility shock	0.248	1.87	−4.910	−2.03	0.07
S&P 500 and VIX change together	0.105	0.73	0.123	2.19	0.36

β on the S&P 500: carry return per 1% S&P move. β on the VIX: carry return (%) per VIX point. FX volatility shock = unexpected change in average daily moves of the eight currencies.

Table 13: Fama–MacBeth test: is carry risk priced? (% per month)

	Estimate	t-statistic
Constant	0.123	0.60
Price of dollar risk	−0.263	−1.06
Price of carry risk	0.233	1.49
Average return of DOL	−0.141	−0.82
Average return of carry factor	0.247	1.83
R^2 across currencies	0.461	–

If the factor is priced correctly, its price of risk should equal its average return. 8 currencies, 119 months.

Priced like a Fama–French factor? Funding currencies have negative carry betas (JPY −0.82, CHF −0.51, EUR −0.32) and high-rate currencies positive ones (AUD 0.48, NOK 0.45, NZD 0.42, CAD 0.31; Annex Table A3). The estimated price of carry risk (0.233% a month) is almost equal to the factor's average return (0.247%), and the two factors explain 46% of cross-currency returns (Table 13). With eight currencies the estimate is imprecise (t = 1.49), so this supports, but does not prove, a priced carry factor.

9 Results: Momentum Filter

Table 14: Carry with and without the three-month momentum filter (December 2016 – July 2026, 116 months, net)

Strategy	Return (% p.a.)	Volatility (% p.a.)	Sharpe	Max drawdown (%)	Skewness	t-stat
Carry	2.52	6.64	0.38	−14.04	−0.51	1.55
Carry + momentum filter	−1.63	7.25	−0.23	−29.00	−0.05	−0.73
Carry and momentum combined rank	−1.80	5.16	−0.35	−28.07	−0.54	−1.09
Momentum only (reference)	−5.41	5.83	−0.93	−44.88	0.43	−2.84

Filter: keep a long only if the currency rose over the last three months, and a short only if it fell. Combined rank: average of the carry rank and the momentum rank. Momentum only: buy the three best and sell the three worst performers of the last three months.

The filter made things worse (Table 14): −1.63% a year (Sharpe −0.23) against +2.52% (Sharpe 0.38) for plain carry, with twice the drawdown. Pure three-month momentum lost 5.41% a year (t = −2.84): major currencies reversed at this horizon, so the filter dropped positions just before they recovered, in line with Menkhoff et al. (2012b). It also left one side empty in a quarter to a third

of months (a hidden dollar bet) and raised turnover from 8% to 62%. It helped slightly in crises (Sharpe 0.04 vs −0.13) but cost far more in calm markets (−0.70 vs 0.64). Volatility-based sizing or option hedges (Jurek, 2014) are better crash protection.

10 Discussion

Risk premium or inefficiency? A risk premium: the return is the rate gap, earned slowly and lost in global sell-offs; returns are left-skewed; carry moves with equities and against the VIX, more so in crises; alpha disappears after risk controls; and the price of carry risk matches its mean. Two cautions: significance is only at 10%, and the result rests on one trade — short CHF, JPY and EUR against a high-yielding dollar. The long basket alone made only 0.13% a year.

Costs and robustness. Raising costs from 0 to 20 bp only lowers the Sharpe from 0.45 to 0.42 (Annex Table A4); FXIP forward spreads (e.g. 15.03/15.09 for EURUSD) imply under 1 bp a month. Policy-rate ranking (Sharpe 0.35) and two-by-two legs (0.40) give similar results.

Capacity. USD, EUR and JPY are the most traded currencies (Bank for International Settlements, 2025), so size is not the limit; crowding is. When many leveraged traders fund in the same currency, as with the yen in 2007 and 2024, the unwind becomes the crash.

Outlook (WIRP). On 17 August 2026 WIRP ranks the currencies like the forwards do (rank correlation 0.98; Annex Table A7). The current portfolio is long AUD, NOK, GBP and short CHF, JPY, CAD. Markets price about 82 bp of BoJ hikes against 38 bp from the Fed, so the yen funding gap is expected to shrink — the set-up behind the 2024 crash.

Limitations. Eight currencies; one deep crisis in the sample; daily returns approximated; fixed costs; WIRP is one snapshot and positioning data were unavailable.

11 Conclusion

A simple G10 carry strategy made 2.9% a year (Sharpe 0.44) in 2016–2026, beating static carry and a dollar basket. It was short CHF, JPY and EUR almost every month and rotated its longs among AUD, NZD, CAD, NOK and GBP. The return was the rate gap: low-rate currencies, especially the yen, did not rise as UIP predicts. The cost is crash risk — left-skewed returns, losses in global shocks and crises, and higher equity correlation under stress. Equity and volatility risk explain most of the return, so carry is a risk premium, not a market error. Momentum filters do not help; sizing and option hedges are better tools. With the BoJ now hiking, the short-yen trade that drove the decade's profits is also its biggest risk.

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Data: Bloomberg L.P. (2026), GP, FRD, WIRP and FXIP functions, accessed August 2026. Event dates: official releases of the Federal Reserve, ECB, Bank of Japan, Bank of England, Swiss National Bank, Norges Bank, RBA, RBNZ, Bank of Canada and Japan's Ministry of Finance.

Annexures

Annex A. Additional results

Table A1: Yearly returns (%)

Year	Carry (net)	Carry + momentum	Static carry	Dollar basket	Months
2016	+3.89	+1.18	+3.89	−5.33	4
2017	+1.61	−11.02	+0.40	+6.05	12
2018	−4.57	−11.40	−3.72	−5.93	12
2019	+2.80	−3.38	+1.40	−0.81	12
2020	−0.71	−2.93	−0.16	+5.41	12
2021	+4.60	−1.89	+3.14	−4.49	12
2022	+0.50	+1.16	+0.61	−8.29	12
2023	+4.09	+5.82	−0.15	−0.78	12
2024	+4.32	−2.10	+0.29	−9.64	12
2025	+3.43	+4.63	+1.54	+6.13	12
2026	+7.60	+3.44	+7.92	+0.13	7

2016 covers September–December; 2026 covers January–July.

Table A2: The ten worst days

Date	Carry (%)	S&P 500 (%)	VIX change	Event
2020-03-09	−3.78	−7.60	12.5	Saudi–Russia oil price war; global stock crash
2020-03-18	−3.57	−5.18	0.5	Rush into US dollars
2025-04-04	−2.86	−5.97	15.3	China retaliates against US tariffs
2020-03-12	−2.38	−9.51	21.6	Stock market crash; ECB disappoints
2021-11-26	−2.31	−2.27	10.0	Omicron variant; oil falls 13%
2025-04-10	−2.17	−3.46	7.1	US–China tariff escalation
2020-03-16	−2.15	−11.98	24.9	Fed cuts to zero, markets still fall
2020-06-11	−1.59	−5.89	13.2	Fed outlook and second-wave fears
2022-05-18	−1.57	−4.04	4.9	US retail earnings shock
2025-04-03	−1.54	−4.84	8.5	Day after US “Liberation Day” tariffs

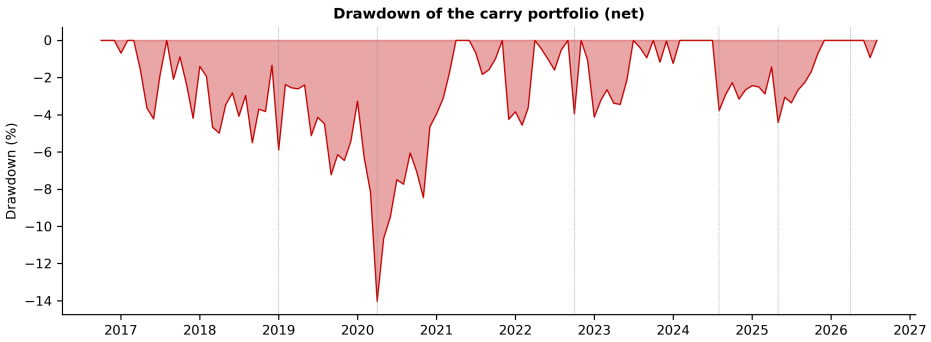


Figure A1: Drawdown of the carry portfolio (net), with the main market shocks marked.

Table A3: Factor betas of each currency (dollar factor and carry factor)

Currency	Beta to dollar factor	Beta to carry factor	Average return (% per month)
EUR	0.92	−0.32	−0.113
GBP	0.88	−0.01	−0.009
JPY	1.04	−0.82	−0.556
CHF	0.97	−0.51	−0.063
NOK	1.28	0.45	−0.100
AUD	1.05	0.48	−0.052
NZD	1.22	0.42	−0.139
CAD	0.65	0.31	−0.092

Table A4: Robustness checks (net)

Version	Return (% p.a.)	Sharpe	Max drawdown (%)
0 bps	2.97	0.449	−14.03
2.5 bps	2.94	0.445	−14.03
5 bps	2.92	0.441	−14.04
10 bps	2.87	0.434	−14.05
20 bps	2.77	0.419	−14.08
Forward-discount signal (baseline)	2.92	0.441	−14.04
Policy-rate differential signal	2.25	0.348	−15.44
Top-2 / bottom-2 (forward signal)	2.91	0.395	−14.49

Table A5: Five largest drawdowns (net, monthly)

Start	Lowest point	Recovered	Depth (%)
2017-08	2020-03	2021-03	−14.04
2021-11	2022-01	2022-03	−4.56
2024-07	2025-04	2025-11	−4.42
2017-03	2017-05	2017-07	−4.22
2022-11	2022-12	2023-06	−4.12

Table A6: Daily statistics

Series	Return (% p.a.)	Volatility (% p.a.)	Sharpe	Max drawdown (%)	Skewness	Excess kurtosis	Worst day (%)
Carry (net)	2.80	6.65	0.42	−18.41	−0.74	8.18	−3.78
Carry + momentum (net)	−1.55	7.17	−0.22	−29.85	−0.42	3.34	−3.42
Dollar basket	−1.63	6.79	−0.24	−26.48	0.02	2.56	−2.74
S&P 500	14.18	18.14	0.78	−33.92	−0.37	15.78	−11.98

Annex B. Python program

The whole analysis is run by one script, `src/carry_project.py`. It reads the Bloomberg files in `data/` and writes every table to `outputs/final/tables/` (also collected in `carry_results.xlsx`) and every chart to `outputs/final/`. The steps are: (1) load spot and one-month forward rates and quote them as USD per currency; (2) take month-end values and compute the forward-discount

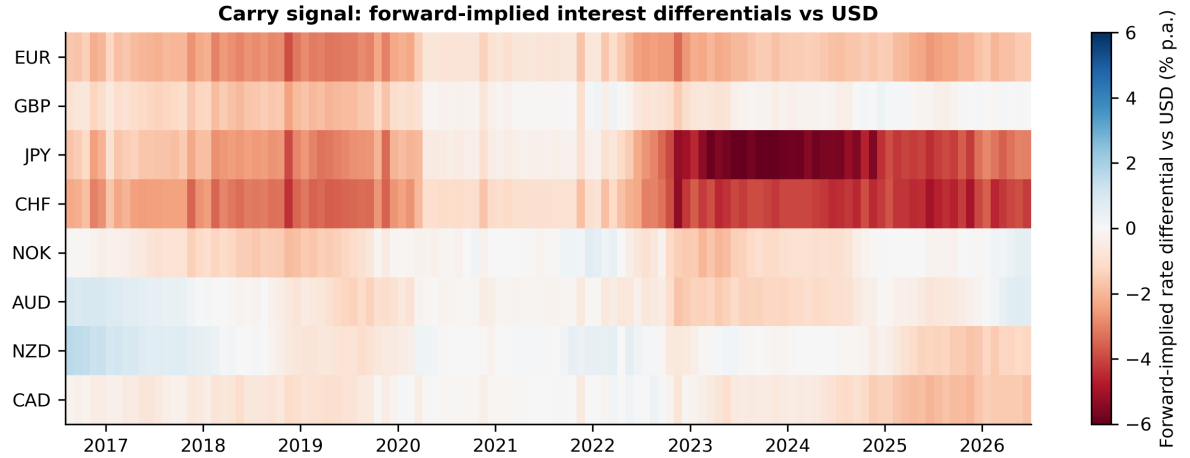


Figure A2: Carry signal over time: forward-implied rate gap of each currency against USD (% p.a.).

Table A7: Bloomberg WIRP snapshot (17 August 2026) as a forward-looking check

Currency	Current rate (%)	Implied rate in about one year (%)	Expected change (bp)	Forward rate gap now (% p.a.)	Rank (WIRP)	Rank (forward)
EUR	2.19	2.78	59	−1.55	6	5
JPY	0.98	1.80	82	−3.39	7	7
CAD	2.28	2.90	62	−1.56	5	6
AUD	4.35	4.79	44	0.64	1	1
NOK	4.25	4.48	23	0.49	2	2
NZD	2.50	3.38	88	−1.12	4	4
CHF	−0.06	0.20	25	−4.28	8	8
GBP	3.73	4.35	62	0.04	3	3

USD: current implied rate 3.63%, about 38 bp of hikes priced to July 2027. Rank correlation between the two rankings: 0.98.

signal; (3) build the carry, momentum, static and dollar-basket portfolios; (4) compute forward-based returns, the spot and carry parts, turnover and costs, and daily returns; (5) performance statistics, sub-periods and VIX regimes; (6) UIP regressions; (7) worst months and days, drawdowns and the tail ratio; (8) correlations, risk regressions and the Fama–MacBeth test; (9) the WIRP check and robustness tests. This paper is built by `src/build_latex.py`, which reads the same results file.

Annex C. Key events

Table A8: Key market and policy events used in this paper

Date	Currencies	Event
2016-11-08	USD, JPY	US election: dollar and US yields rise, yen falls
2016-12-14	USD	Fed hikes to 0.50–0.75%; start of the 2017–18 hiking cycle
2018-02-05	All	VIX spike ("Volmageddon"); carry −2.8% in February 2018
2018-07-06	AUD, NZD	First US tariffs on China; trade war hits AUD and NZD
2018-12-19	USD	Fed hikes to 2.25–2.50%; late-2018 sell-off
2019-01-03	JPY, AUD	Yen flash crash: AUD/JPY falls about 7% in minutes
2019-08-05	JPY, AUD	Chinese yuan breaks 7 per dollar; US calls China a currency manipulator
2020-03-09	NOK, CAD	Saudi–Russia oil price war; stock crash
2020-03-15	All	Fed cuts to zero; dollar swap lines to other central banks
2020-11-09	All	Vaccine news; risk rally (carry +4.2% in November 2020)
2021-11-26	NOK, CAD	Omicron variant; oil sell-off
2022-02-24	EUR, NOK	Russia invades Ukraine; energy shock
2022-09-21/23	USD, GBP, JPY	Fed +75 bp; UK mini-budget; Japan buys yen (22 Sep)
2022-11-10	USD	Soft US inflation data: largest one-day dollar fall since 2009
2022-12-20	JPY	Bank of Japan widens its bond-yield band; yen jumps
2023-03-10/19	CHF, JPY	Silicon Valley Bank fails; UBS takes over Credit Suisse
2024-03-19	JPY	Bank of Japan ends negative rates
2024-07-11/31	JPY	Japan buys yen; BoJ hikes to 0.25%; yen-carry unwind starts
2024-08-05	JPY, all	Global carry crash: Nikkei −12.4%
2025-04-02	All	US "Liberation Day" tariffs; 90-day pause on 9 April
2025-12-19	JPY	Bank of Japan hikes to 0.75%
2026-02-28	NOK, CAD, JPY	US–Israel strikes on Iran; Strait of Hormuz closed; oil above $100
2026-02 to 05	AUD	RBA hikes three times, 3.60% to 4.35%
2026-04-28	JPY	Japan starts record yen buying (¥11.7tn, to 27 May)
2026-06-11/16	EUR, JPY	ECB hikes to 2.25%; Bank of Japan hikes to 1.00%

Full list of about 370 events: `data/macro_event_log.xlsx`.

Annex D. Bloomberg functions

Table A9: Bloomberg functions used

Function	What it gives	How it is used
GP	Daily spot rates for the 8 pairs; S&P 500 and VIX	All returns, regimes and correlations
FRD	One-month forward points	Carry signal and forward-based returns
WIRP	Market-implied central-bank rate paths (17 Aug 2026)	Forward-looking check (Table A7)
FXIP	Forward points, overnight to six months (bid/ask)	Roll-cost estimate; positioning data not available
CORR	Rolling correlation of carry with SPX and VIX	Check Figure 9 on the terminal (252-day window)
REGR	Regression of carry on SPX and VIX change	Check Table 12 on the terminal
BTST	Strategy backtest	Optional check of Table 4

Bloomberg Screenshots

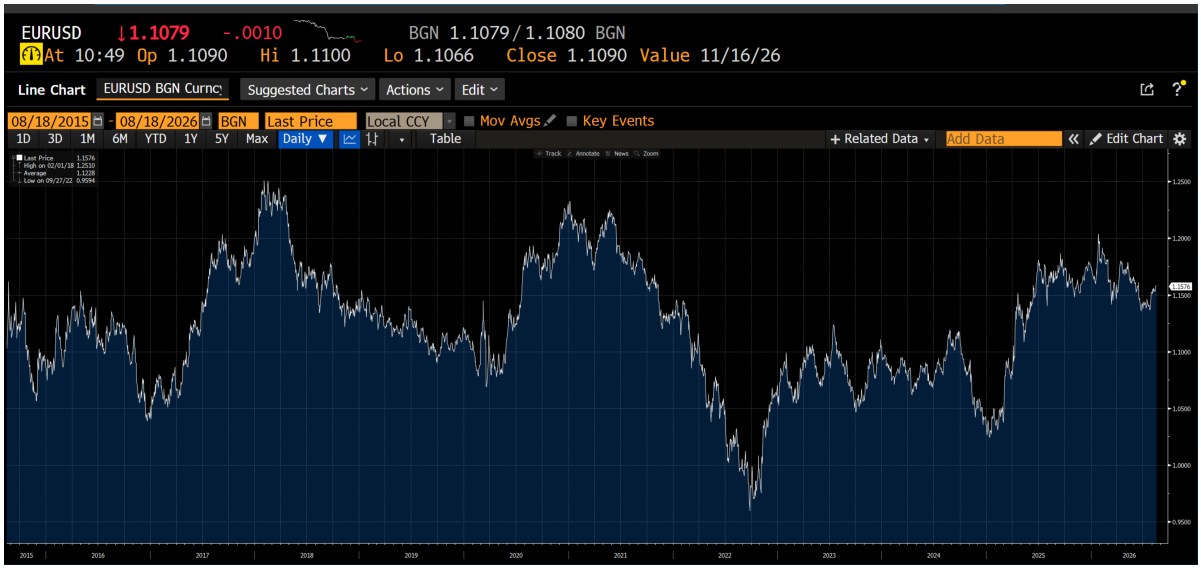


Figure A3: Bloomberg GP: EURUSD daily spot, Aug 2015–Aug 2026. Source of the daily spot series (S) driving the carry signal and portfolio returns throughout the pipeline.



Figure A4: Bloomberg FRD: EURUSD 1-month forward points. Source of the forward points added to spot to build F_t, used in the carry signal $c = S/F - 1$.

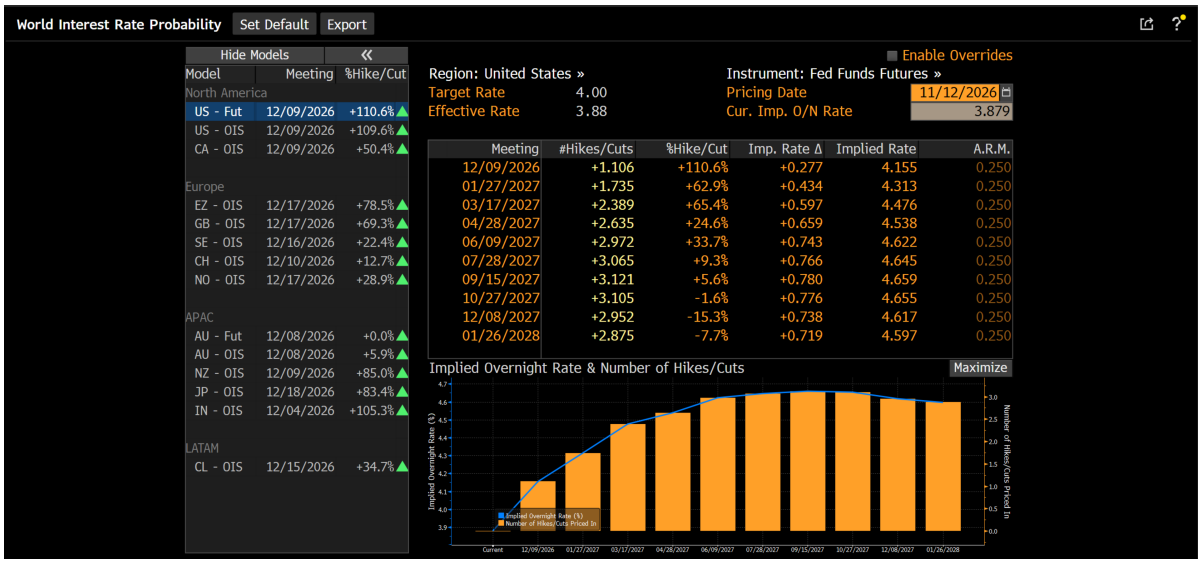


Figure A5: Bloomberg WIRP: market-implied Fed rate path, priced 11/12/2026. Forward-looking policy-rate expectations used for the WIRP robustness check against the forward-implied carry signal (Table A7).

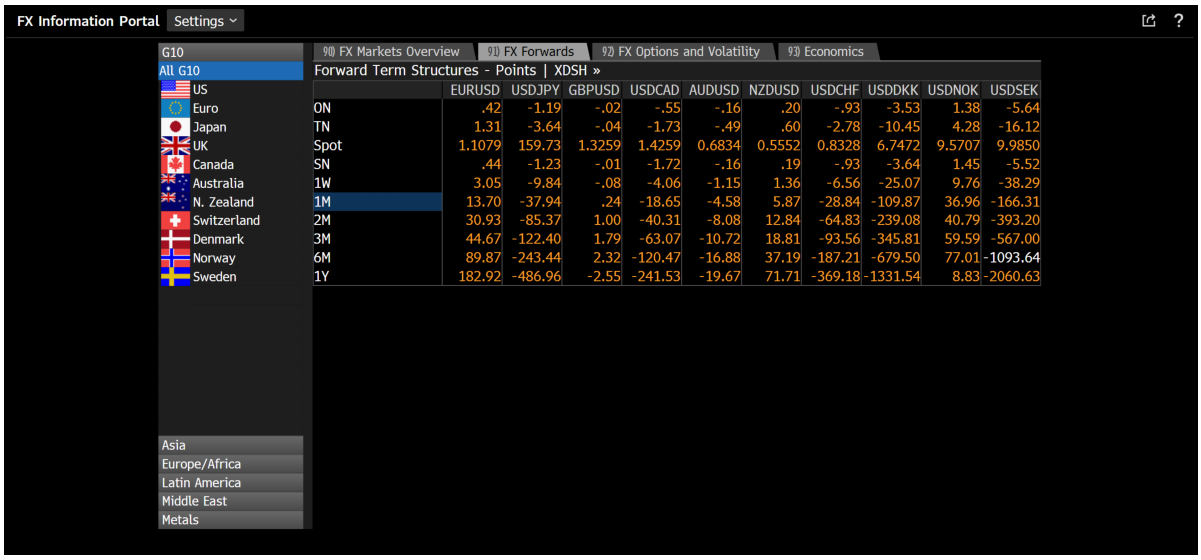


Figure A6: Bloomberg FXIP: G10 forward points across tenors. Cross-check confirming the 1-month forward points used in the pipeline are consistent with the broader forward curve across all eight pairs.

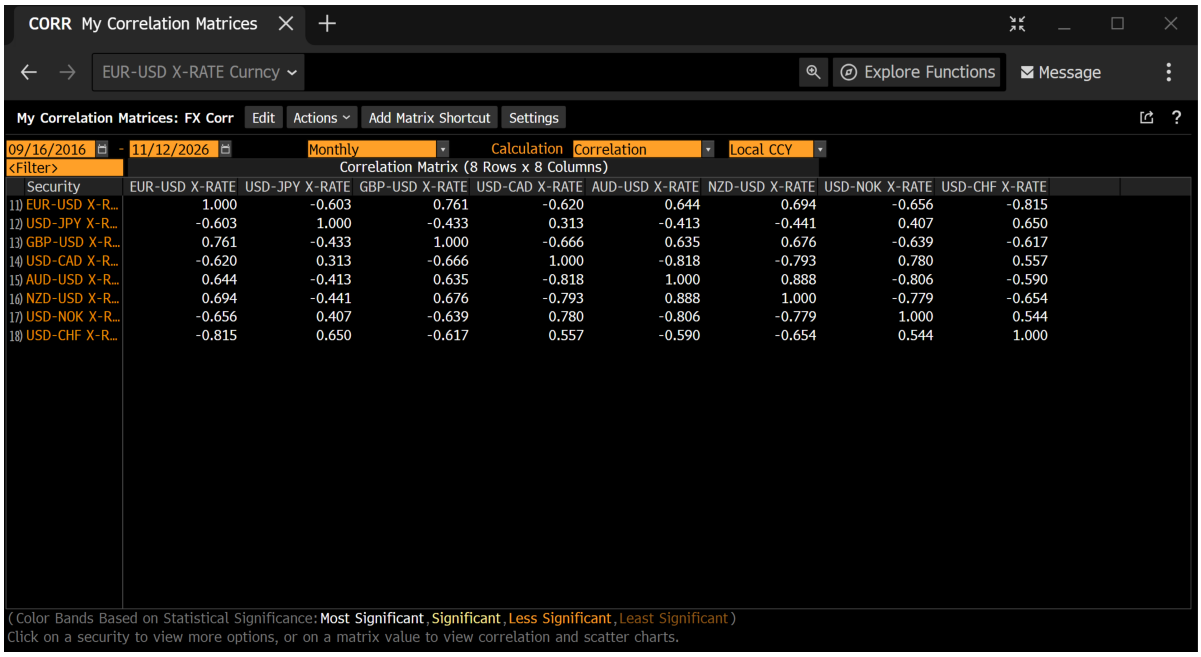


Figure A7: Bloomberg CORR: correlation matrix of G10 spot returns. Terminal-side confirmation of the cross-currency correlations underlying the portfolio's diversification assumptions.



Figure A8: Bloomberg HRA: EURUSD regressed on the S&P 500. Terminal-side cross-check of the equity-exposure coefficient reported in Table 12.

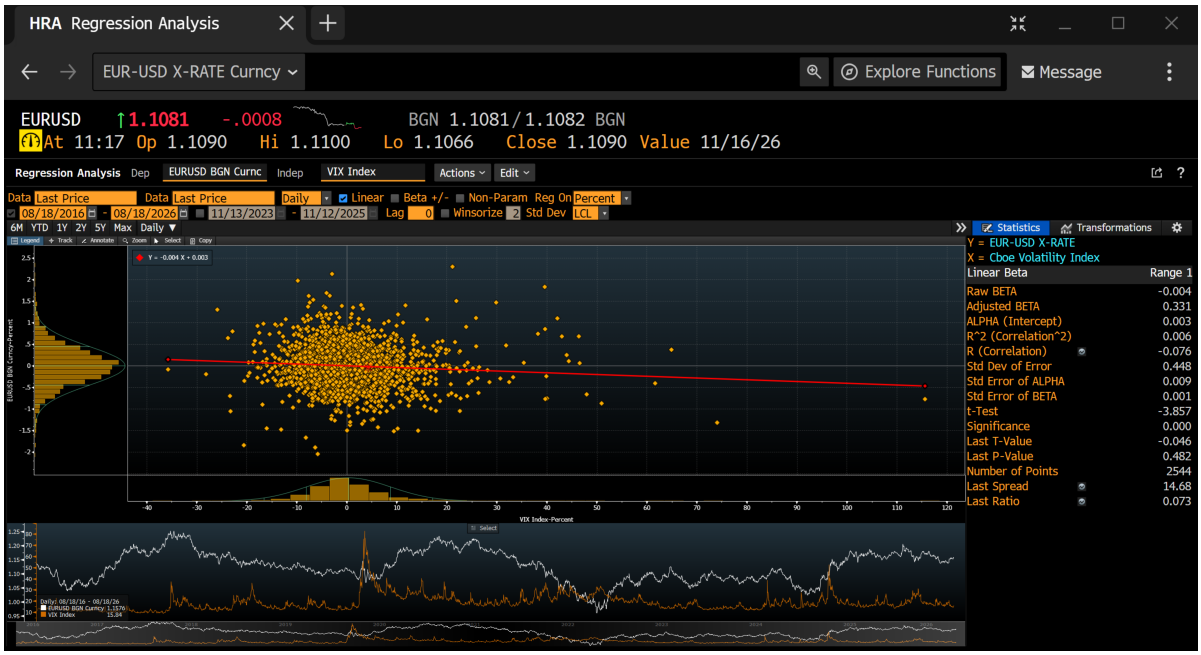


Figure A9: Bloomberg HRA: EURUSD regressed on the VIX. Terminal-side cross-check of the volatility-exposure coefficient reported in Table 12.

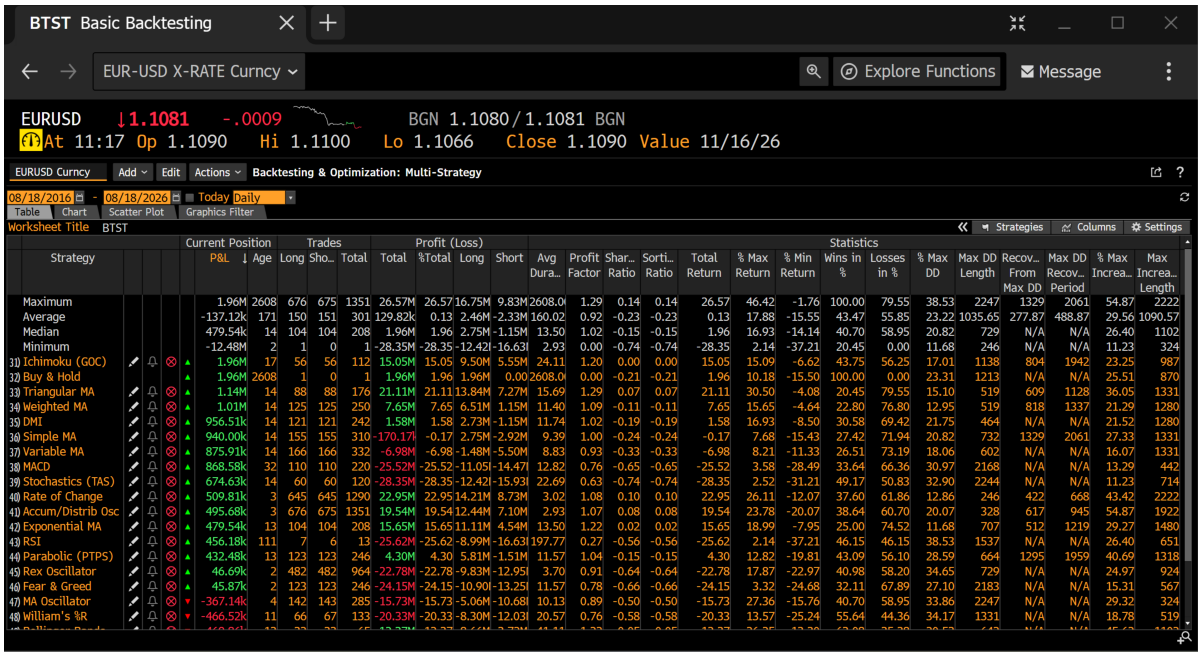


Figure A10: Bloomberg BTST: multi-strategy backtest on EURUSD (table view). High-level terminal backtest shown as a sanity check against the Python-computed performance figures in Table 4 — not the primary results source.

Code Snippets

load_fx() — currency conversion to USD-per-foreign convention

```python
def load_fx():
    spot, fwd = {}, {}
    for c in CCY:
        df = pd.read_excel(DATA / "forward_series_1m.xlsx", sheet_name=SHEET[c])
        df["date"] = pd.to_datetime(df["date"], errors="coerce")
        for k in ("spot", "forward_1m"):
            df[k] = pd.to_numeric(df[k], errors="coerce")
        df = (df.dropna(subset=["date", "spot", "forward_1m"])
                .drop_duplicates("date", keep="last").set_index("date").sort_index())
        s, f = df["spot"], df["forward_1m"]
        if c not in DIRECT:                    # invert JPY/CHF/CAD/NOK to USD-per-foreign
            s, f = 1.0 / s, 1.0 / f
        spot[c], fwd[c] = s, f
    S = pd.DataFrame(spot).sort_index().ffill(limit=3).dropna()
    F = pd.DataFrame(fwd).reindex(S.index).ffill(limit=3)
    ok = F.notna().all(axis=1)
    return S[ok], F[ok]
```

Figure A11: `load_fx()` — converts USDJPY, USDCHF, USDCAD and USDNOK into the same "USD per unit of foreign currency" convention as the other four pairs, so all eight series are comparable.

Carry signal and excess-return construction (no look-ahead)

```python
carry_sig = S_m / F_m - 1                          # c_t: 1M forward discount signal
diff_ann = carry_sig * 1200                         # annualized % (approx i_for - i_usd)

# Holding-month returns: label = holding month m; formation = m-1
x_ex    = (S_m / F_m.shift(1) - 1).iloc[1:]         # excess return  x_{t+1} = S_{t+1}/F_t - 1
x_spot  = (S_m / S_m.shift(1) - 1).iloc[1:]         # spot component
x_carry = carry_sig.shift(1).iloc[1:]               # carry component (known at t)
sig_form = carry_sig.shift(1).iloc[1:]              # signal used for holding month m
```

Figure A12: Computes the forward-discount signal and the realized excess return, with every signal lagged by one month so no future information leaks into portfolio formation.

carry_weights() + run_portfolio() — portfolio construction & costs

```python
def carry_weights(sig: pd.DataFrame, n=N_LEG) -> pd.DataFrame:
    W = pd.DataFrame(0.0, index=sig.index, columns=CCY)
    for m, row in sig.iterrows():
        r = row.dropna()
        W.loc[m, r.nlargest(n).index] = 1.0 / n       # top-3  -> long
        W.loc[m, r.nsmallest(n).index] = -1.0 / n     # bottom-3 -> short
    return W


def run_portfolio(W: pd.DataFrame, tc_bps=TC_BPS):
    W = W.reindex(months).fillna(0.0)
    gross = (W * x_ex).sum(axis=1)
    turnover = 0.5 * W.diff().abs().sum(axis=1)       # 0.5x: a reversal counts as one trade
    turnover.iloc[0] = 0.5 * W.iloc[0].abs().sum()
    cost = turnover * tc_bps / 1e4
    return pd.DataFrame({"gross": gross, "net": gross - cost,
                         "spot": (W * x_spot).sum(axis=1),
                         "carry": (W * x_carry).sum(axis=1),
                         "turnover": turnover, "cost": cost})
```

Figure A13: `carry_weights()` and `run_portfolio()` — ranks currencies by the signal, assigns $+1/3$ to the top three (long) and $-1/3$ to the bottom three (short), and deducts a 5 bp transaction cost per unit of turnover.

perf() — Sharpe, Sortino, CAGR, Calmar, tail ratio, NW t-stat

```python
def perf(r: pd.Series, freq=12) -> dict:
    r = r.dropna(); n = len(r)
    mu, sd = r.mean(), r.std(ddof=1)
    dd_min, _ = max_dd(r)
    cagr = (1 + r).prod() ** (freq / n) - 1
    down = np.sqrt((np.minimum(r, 0) ** 2).mean())
    t_nw = hac(r.values, np.ones(n)).tvalues[0]
    k = max(1, int(np.ceil(0.05 * n)))
    srt = np.sort(r.values)
    return {"Mean ann. (%)": mu * freq * 100,
            "CAGR (%)": cagr * 100,
            "Vol ann. (%)": sd * np.sqrt(freq) * 100,
            "Sharpe": mu / sd * np.sqrt(freq),
            "Sortino": mu / down * np.sqrt(freq) if down > 0 else np.nan,
            "Max drawdown (%)": dd_min * 100,
            "Calmar": cagr / abs(dd_min) if dd_min < 0 else np.nan,
            "Tail ratio": abs(srt[:k].mean()) / srt[-k:].mean(),
            "NW t-stat (mean=0)": t_nw}
```

Figure A14: `perf()` — computes every headline risk/return statistic reported in Table 4 (Sharpe, Sortino, CAGR, Calmar, tail ratio) from one monthly return series.

UIP / Fama (1984) regression, per currency

```python
ds = np.log(S_m).diff().iloc[1:] * 1200                    # % p.a., + = foreign appreciation
fp = (np.log(F_m) - np.log(S_m)).shift(1).iloc[1:] * 1200   # f_t - s_t (known at t), % p.a.

uip_rows = {}
for c in CCY:
    y, xv = ds[c], fp[c]
    m_ = hac(y.values, sm.add_constant(xv.values))          # Delta s = alpha + beta*(f-s) + eps
    b, se = m_.params[1], m_.bse[1]
    uip_rows[c] = {"beta": b, "HAC s.e.": se,
                   "t (beta=1)": (b - 1) / se,              # tests UIP null: beta = 1
                   "p (beta=1)": 2 * stats.norm.sf(abs((b - 1) / se)),
                   "R2": m_.rsquared}
```

Figure A15: Runs the Fama (1984) regression of next-month spot change on the current forward discount, per currency, to test whether UIP ($\beta = 1$) holds.

Position-history table (Table 3 source)

```python
# Position history: collapse consecutive months with the same long/short basket
_l = [", ".join(sorted(W_carry.columns[W_carry.loc[m] > 0])) for m in months]
_s = [", ".join(sorted(W_carry.columns[W_carry.loc[m] < 0])) for m in months]
_hist, _start = [], 0
for i in range(1, len(months) + 1):
    if i == len(months) or (_l[i], _s[i]) != (_l[_start], _s[_start]):
        seg = P["net"].iloc[_start:i]
        _hist.append({"From": str(months[_start]), "To": str(months[i - 1]),
                      "Months": i - _start,
                      "Long (high-yield)": _l[_start], "Short (funding)": _s[_start],
                      "Return over period (%)": ((1 + seg).prod() - 1) * 100})
        _start = i
```

Figure A16: Collapses consecutive months with an unchanged long/short basket into single rows, producing the position-history table (Table 3).